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Maths Selected Questions for JEE Mains

Sequence and Series

(12 Questions)

$(20)^{19} + 2(21)(20)^{18} + 3(21)^2(20)^{17} + \dots + 20(21)^{19} = k(20)^{19}$,
then k is equal to _____.

For three positive integers p, q, r , $x^{pq^2} = y^{qr} = z^{p^2r}$ and $r = pq + 1$ such that $3, 3 \log_y x, 3 \log_z y, 7 \log_x z$ are in A.P. with common difference $\frac{1}{2}$. Then $r-p-q$ is equal to

A 12

C 6

B -6

D 2

Let A_1 and A_2 be two arithmetic means and G_1, G_2, G_3 be three geometric means of two distinct positive numbers. Then $G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$ is equal to :

A $(A_1 + A_2)^2 G_1 G_3$

C $2(A_1 + A_2)G_1^2 G_3^2$

B $(A_1 + A_2)G_1^2 G_3^2$

D $2(A_1 + A_2)G_1 G_3$

The sum of the common terms of the following three arithmetic progressions.

3, 7, 11, 15, ..., 399,

2, 5, 8, 11, ..., 359 and

2, 7, 12, 17, ..., 197,

is equal to _____.

Let $\sum_{n=0}^{\infty} \frac{n^3((2n)!)+(2n-1)(n!)}{(n!)((2n)!)} = ae + \frac{b}{e} + c$, where $a, b, c \in \mathbb{Z}$ and $e = \sum_{n=0}^{\infty} \frac{1}{n!}$. Then $a^2 - b + c$

is equal to_____

Let $0 < z < y < x$ be three real numbers such that $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in an arithmetic progression and $x, \sqrt{2}y, z$ are in a geometric progression. If $xy + yz + zx = \frac{3}{\sqrt{2}}xyz$, then $3(x + y + z)^2$ is equal to_____.

Let $\langle a_n \rangle$ be a sequence such that $a_1 + a_2 + \dots + a_n = \frac{n^2+3n}{(n+1)(n+2)}$. If $28 \sum_{k=1}^{10} \frac{1}{a_k} = p_1 p_2 p_3 \dots p_m$, where $p_1, p_2, \dots, p_m$ are the first m prime numbers, then m is equal to

A 5

C 6

B 7

D 8

Let $a_1, a_2, \dots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then

$12 \left(\frac{1}{\sqrt{a_{10}} + \sqrt{a_{11}}} + \frac{1}{\sqrt{a_{11}} + \sqrt{a_{12}}} + \dots + \frac{1}{\sqrt{a_{17}} + \sqrt{a_{18}}} \right)$ is equal to .

Let $a_1 = b_1 = 1$ and $a_n = a_{n-1} + (n-1), b_n = b_{n-1} + a_{n-1}, \forall n \geq 2$. If $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ and $T = \sum_{n=1}^8 \frac{n}{2^{n-1}}$, then $2^7(2S - T)$ is equal to _____.

Suppose a_1, a_2, a_3, a_4 be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $49/2$, then a_4 is equal to _____.

Let $\{a_k\}$ and $\{b_k\}, k \in N$, be two G.P.s with common ratios r_1 and r_2 respectively such that $a_1 = b_1 = 4$ and $r_1 < r_2$. Let $c_k = a_k + b_k, k \in N$. If $c_2 = 5$ and $c_3 = \frac{13}{4}$ then $\sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$ is equal to _____.

The sum of 10 terms of the series

$$\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots \text{ is}$$

A $\frac{58}{111}$

C $\frac{55}{111}$

B $\frac{56}{111}$

D $\frac{59}{111}$

Quadratic Equation

(8 Questions)

The number of integral values of k , for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other root lies in the interval $(2, 3)$, is :

A 2

C 1

B 0

D 3

Let $\alpha_1, \alpha_2, \dots, \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1| \geq |\alpha_2| \geq \dots \geq |\alpha_7|$. Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____.

The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0, x \in \mathbb{R}$ has :

A two solutions and both are negative

C four solutions two of which are negative

B two solutions and only one of them is negative

D no solution

Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then $\frac{\alpha^{23} + \beta^{23} + \alpha^{14} + \beta^{14}}{\alpha^{15} + \beta^{15} + \alpha^{10} + \beta^{10}}$ is equal

A 72

C 729

B 9

D 81

Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation

☐ A $7x^2 - 245x + 250 = 0$

☐ C $49x^2 + 245x + 250 = 0$

☐ B $49x^2 - 245x + 250 = 0$

☐ D $7x^2 + 245x - 250 = 0$

Let $S = \left\{ x : x \in \mathbb{R} \text{ and } (\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10 \right\}$. Then $n(S)$ is equal to

☐ A 6

☐ C 0

☐ B 4

☐ D 2

The set of all $a \in \mathbb{R}$ for which the equation $x|x - 1| + |x + 2| + a = 0$ has exactly one real root, is :

☐ A $(-\infty, \infty)$

☐ C $(-\infty, -3)$

☐ B $(-6, \infty)$

☐ D $(-6, -3)$

Let $\lambda \in \mathbb{R}$ and let the equation E be $|x|^2 - 2|x| + |\lambda - 3| = 0$. Then the largest element in the set $S = \{ x + \lambda : x \text{ is an integer solution of E} \}$ is _____

Integration

(15 Questions)

The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x \left(\log_e \left(e^x + \sqrt{1 + e^{2x}} \right) \right) dx$ is equal to :

A $\log_e \left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$

C $\log_e \left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$

B $\log_e \left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) - \frac{\sqrt{5}}{2}$

D $\log_e \left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}} \right) + \frac{\sqrt{5}}{2}$

$\lim_{n \rightarrow \infty} \left\{ \left(2^{\frac{1}{2}} - 2^{\frac{1}{3}} \right) \left(2^{\frac{1}{2}} - 2^{\frac{1}{5}} \right) \dots \left(2^{\frac{1}{2}} - 2^{\frac{1}{2n+1}} \right) \right\}$ is equal to :

A $\sqrt{2}$

C $\frac{1}{\sqrt{2}}$

B 1

D 0

Let $[t]$ denote the greatest integer $\leq t$. Then $\frac{2}{\pi} \int_{\pi/6}^{5\pi/6} (8[\operatorname{cosec} x] - 5[\cot x]) dx$

is equal to _____

If $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $\int_0^{\frac{\pi}{2}} f(\sin 2x) \sin x dx + \alpha \int_0^{\frac{\pi}{4}} f(\cos 2x) \cos x dx = 0$, then the value of α is :

A $-\sqrt{3}$

C $-\sqrt{2}$

B $\sqrt{2}$

D $\sqrt{3}$

Let the function $f : [0, 2] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} e^{\min\{x^2, x - [x]\}}, & x \in [0, 1) \\ e^{[x - \log_e x]}, & x \in [1, 2] \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t . Then the value of the integral $\int_0^2 x f(x) dx$ is :

A $2e - 1$

C $1 + \frac{3e}{2}$

B $2e - \frac{1}{2}$

D $(e - 1) \left(e^2 + \frac{1}{2} \right)$

If $\int_0^1 (x^{2l} + x^{14} + x^7) (2x^{14} + 3x^7 + 6)^{1/7} dx = \frac{1}{l} (11)^{m/n}$ where $l, m, n \in \mathbb{N}$, m and n are coprime then $l + m + n$ is equal to _____.

Let $f(x) = \frac{x}{(1+x^n)^{\frac{1}{n}}}$, $x \in \mathbb{R} - \{-1\}$, $n \in \mathbb{N}$, $n > 2$.

If $f^n(x) = (f \circ f \circ f \dots \text{upto } n \text{ times})(x)$, then

$\lim_{n \rightarrow \infty} \int_0^1 x^{n-2} (f^n(x)) dx$ is equal to _____.

Let $\alpha \in (0, 1)$ and $\beta = \log_e(1 - \alpha)$. Let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n}$, $x \in (0, 1)$. Then the integral $\int_0^\alpha \frac{t^{50}}{1-t} dt$ is equal to

A $-(\beta + P_{50}(\alpha))$

C $P_{50}(\alpha) - \beta$

B $\beta - P_{50}(\alpha)$

D $\beta + P_{50} - (\alpha)$

The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3 \sin x)}{\sin x(1+\cos x)} dx$ is equal to :

A $\frac{10}{3} - \sqrt{3} + \log_e \sqrt{3}$

C $\frac{10}{3} - \sqrt{3} - \log_e \sqrt{3}$

B $\frac{7}{2} - \sqrt{3} - \log_e \sqrt{3}$

D $-2 + 3\sqrt{3} + \log_e \sqrt{3}$

Let $f(x) = x + \frac{a}{\pi^2-4} \sin x + \frac{b}{\pi^2-4} \cos x$, $x \in R$ be a function which

satisfies $f(x) = x + \int_0^{\pi/2} \sin(x+y) f(y) dy$. then $(a+b)$ is equal to

A $-2\pi(\pi+2)$

C $-\pi(\pi+2)$

B $-\pi(\pi-2)$

D $-2\pi(\pi-2)$

The value of the integral $\int_{1/2}^2 \frac{\tan^{-1}x}{x} dx$ is equal to :

A $\frac{\pi}{2} \log_e 2$

C $\frac{1}{2} \log_e 2$

B $\frac{\pi}{4} \log_e 2$

D $\pi \log_e 2$

The minimum value of the function $f(x) = \int_0^2 e^{|x-t|} dt$ is :

A 2

C $e(e - 1)$

B $2(e - 1)$

D $2e - 1$

The value of $\frac{e^{-\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x} \tan^{50} x dx}{\int_0^{\frac{\pi}{4}} e^{-x} (\tan^{49} x + \tan^{51} x) dx}$ is

A 51

C 25

B 50

D 49

For $m, n > 0$, let $\alpha(m, n) = \int_0^2 t^m (1 + 3t)^n dt$. If $11\alpha(10, 6) + 18\alpha(11, 5) = p(14)^6$,

then p is equal to_____

Let $\alpha \in (0, 1)$ and $\beta = \log_e(1 - \alpha)$. Let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n}$, $x \in (0, 1)$. Then the integral $\int_0^\alpha \frac{t^{50}}{1-t} dt$ is equal to

☐ A $-(\beta + P_{50}(\alpha))$

☐ C $P_{50}(\alpha) - \beta$

☐ B $\beta - P_{50}(\alpha)$

☐ D $\beta + P_{50} - (\alpha)$
